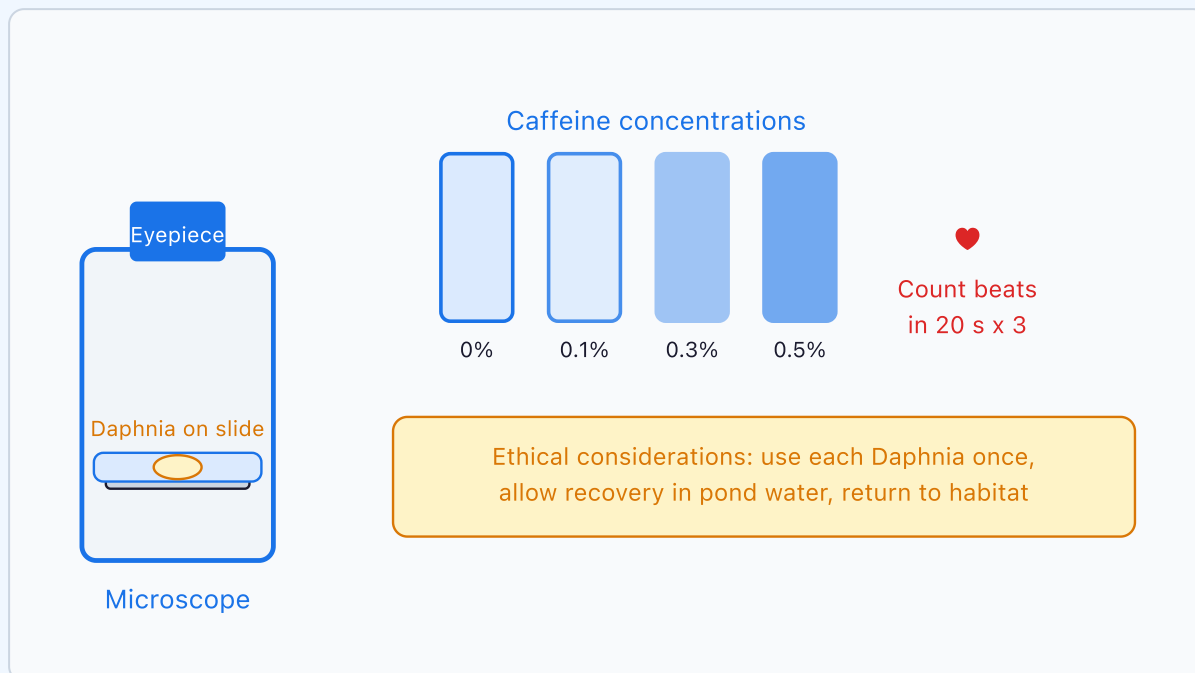


Edexcel A (Salters-Nuffield) A-Level Biology — Core Practicals

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CP1: Effect of Caffeine on Daphnia Heart Rate



Background Theory

Daphnia (water fleas) are small crustaceans with transparent bodies, making their heart visible under a microscope without dissection. Their heart is myogenic (contracts without nervous stimulation) and typically beats at 200–300 bpm at room temperature. **Caffeine** is a stimulant drug that inhibits **phosphodiesterase**, the enzyme that breaks down cAMP (cyclic AMP). This causes cAMP to accumulate, which keeps protein kinase A active for longer, increasing the rate and force of heart contractions. This is the same mechanism by which caffeine increases heart rate in humans.

Apparatus

- Live *Daphnia* (water fleas), cavity slides, cotton wool strands (to restrict movement without crushing)
- Microscope (low power only — high power concentrates too much light/heat), dropping pipette
- Caffeine solutions at a range of concentrations (e.g. 0%, 0.1%, 0.3%, 0.5% w/v), prepared by serial dilution
- Pond water or dechlorinated tap water (for control and recovery), stopwatch

Method

1. Transfer a single *Daphnia* using a wide-bore pipette onto a cavity slide in a small drop of pond water. Place a few cotton wool strands over it to restrict movement without applying pressure.
2. View under low-power microscope. Locate the heart (visible as a pulsating structure in the dorsal part of the transparent body). Allow 2 minutes for the organism to acclimatise.
3. Count heartbeats for **20 seconds**. Multiply by 3 to obtain beats per minute (bpm). Repeat the 20 s count three times and calculate the mean. Using 20 s intervals rather than 60 s reduces the time the *Daphnia* spends on the slide.
4. Add 2–3 drops of the lowest caffeine concentration to the slide using a pipette. Wait **2 minutes** for the caffeine to diffuse into the organism and take effect.
5. Count heartbeats again (three 20 s counts). Record and calculate mean bpm.
6. Use a **fresh *Daphnia*** for each concentration to avoid cumulative drug effects. Return all *Daphnia* to pond water promptly after use.

Key Variables

Variable	Detail
Independent	Caffeine concentration (0%, 0.1%, 0.3%, 0.5%)
Dependent	Heart rate (beats per minute)
Control variables	Temperature (use LED light to minimise heating), acclimatisation time, volume of solution added, size of <i>Daphnia</i> (select similar-sized individuals), same microscope light intensity

Safety and Ethical Considerations

- **Ethical treatment:** Use the minimum number of organisms necessary; handle gently with wide-bore pipettes; minimise time on the slide; return to pond water immediately after experiment; use low concentrations of caffeine that are sub-lethal
- Use **LED or fibre-optic illumination** (not incandescent bulbs) to reduce heat transfer to the *Daphnia*. A traditional microscope lamp can raise the temperature of the slide by several degrees, independently increasing heart rate

- Each Daphnia should only be used **once** to avoid cumulative/sensitisation effects and ethical concerns about repeated stress
- Caffeine solutions at these concentrations are low hazard to humans

Analysis

- Calculate mean heart rate (bpm) for each caffeine concentration. Include standard deviation or range as a measure of spread.
- Plot a graph of mean heart rate (y-axis) against caffeine concentration (x-axis). Expect a positive correlation: higher caffeine concentration → higher heart rate.
- Explain using pharmacology: caffeine inhibits phosphodiesterase → cAMP not broken down → cAMP-dependent protein kinase remains active → increased rate of SA node depolarisation → increased heart rate.
- Statistical test: use a Spearman's rank correlation (if testing for correlation) or a Mann-Whitney U test (if comparing two specific groups).

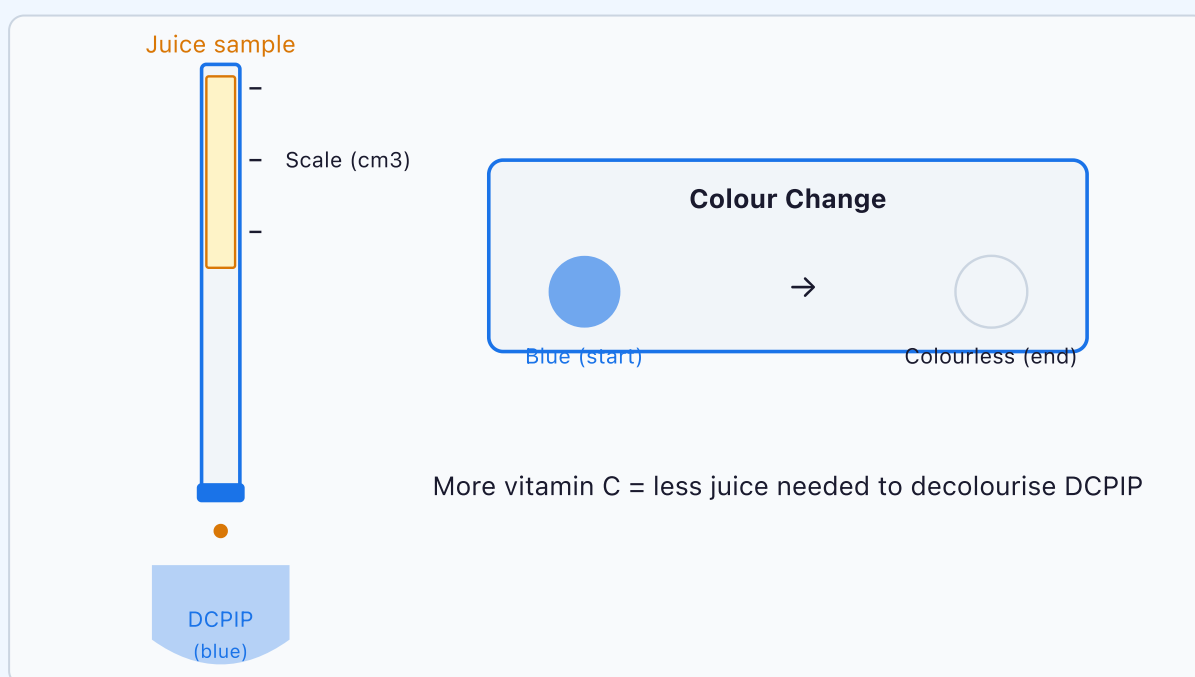
Sources of Error and Improvements

Error	Effect	Improvement
Heart beats too fast to count accurately by eye	Likely underestimate; inconsistency between observers	Use the "tapping" method: tap a pencil on paper in time with heartbeats, then count marks. Alternatively, record video through the microscope and count in slow-motion playback
Heat from microscope lamp	Raises temperature, independently increasing heart rate (confounding variable)	Use LED or fibre-optic light source; limit time under lamp; place a heat filter (beaker of water) between light and stage
Stress from handling and being on a slide	Elevated baseline heart rate due to stress response	Allow 2–3 minutes acclimatisation before first count; handle gently; use wide-bore pipette

Same Daphnia reused for multiple concentrations	Cumulative effect of caffeine; no washout period; ethical concerns	Use a fresh Daphnia for each concentration; randomise order of testing
Caffeine concentration may not be uniform on slide	Daphnia may not be exposed to the intended concentration	Gently mix after adding caffeine drops; allow sufficient diffusion time

Exam Tip: This practical is commonly used to test ethical considerations in animal experiments. Always discuss: (1) using the minimum number of organisms, (2) using the lowest effective concentrations, (3) providing recovery time, (4) returning organisms to their habitat, (5) why Daphnia are an appropriate model (transparent body, visible heart, invertebrate so fewer ethical restrictions than vertebrates). Also be prepared to explain why a fresh Daphnia is needed for each concentration — to avoid cumulative drug effects that would confound results.

CP2: Vitamin C Content (DCPIP Titration)



Background Theory

DCPIP (2,6-dichlorophenolindophenol) is a blue dye that acts as an indicator for reducing agents. **Vitamin C (ascorbic acid)** is a powerful reducing agent (antioxidant) that donates electrons to DCPIP, reducing it from its oxidised blue form to its reduced colourless form. The volume of juice required to completely decolourise a fixed volume of DCPIP is inversely proportional to the vitamin C concentration — a higher vitamin C content means less juice is needed.

Apparatus

- DCPIP solution (0.1%), test tubes, test tube rack
- Fruit juice samples (e.g. fresh orange, pasteurised orange, lemon, apple, blackcurrant)
- Graduated pipette (1 cm³) or burette for accurate volume measurement, syringe
- Known concentrations of ascorbic acid solution (e.g. 0.02%, 0.04%, 0.06%, 0.08%, 0.1% w/v) for constructing a calibration curve
- White tile (to judge colour change against a white background)

Method

1. Pipette exactly **1 cm³** of DCPIP solution into a test tube.
2. Using a graduated pipette or burette, add the juice sample **drop by drop**, shaking/swirling gently after each addition. Place the tube against a white tile to judge colour.
3. Record the exact volume of juice needed to **completely decolourise** the DCPIP (blue → colourless). This is the end-point.
4. Repeat with each juice sample. Perform at least **3 repeats** per sample and calculate the mean volume.
5. Construct a **calibration curve** by repeating the procedure using known concentrations of ascorbic acid solution instead of juice. Plot volume needed to decolourise (x-axis) against known concentration (y-axis).
6. Use the calibration curve to determine the vitamin C concentration of each unknown juice sample from its mean volume.

Key Variables

Variable	Detail
Independent	Type of juice (or storage method: fresh vs pasteurised vs boiled)
Dependent	Volume of juice needed to decolourise 1 cm ³ DCPIP
Control variables	Volume and concentration of DCPIP, temperature, rate of addition, same batch of DCPIP

Safety

- DCPIP stains skin and clothing — wear gloves and a lab coat.
- Eye protection when handling solutions.
- Ascorbic acid is a mild irritant at high concentrations — low risk at concentrations used.

Analysis

- **Less juice needed** to decolourise = **higher vitamin C content**.
- Plot a calibration curve from known ascorbic acid concentrations. Read off the concentration for each juice from the curve using interpolation.
- Compare different juices or investigate how processing (heating, storage, exposure to air/light) affects vitamin C content.
- Vitamin C is heat-labile and light-sensitive, so pasteurised juice typically has less than fresh juice, and juice stored in clear containers loses more than juice in opaque containers.

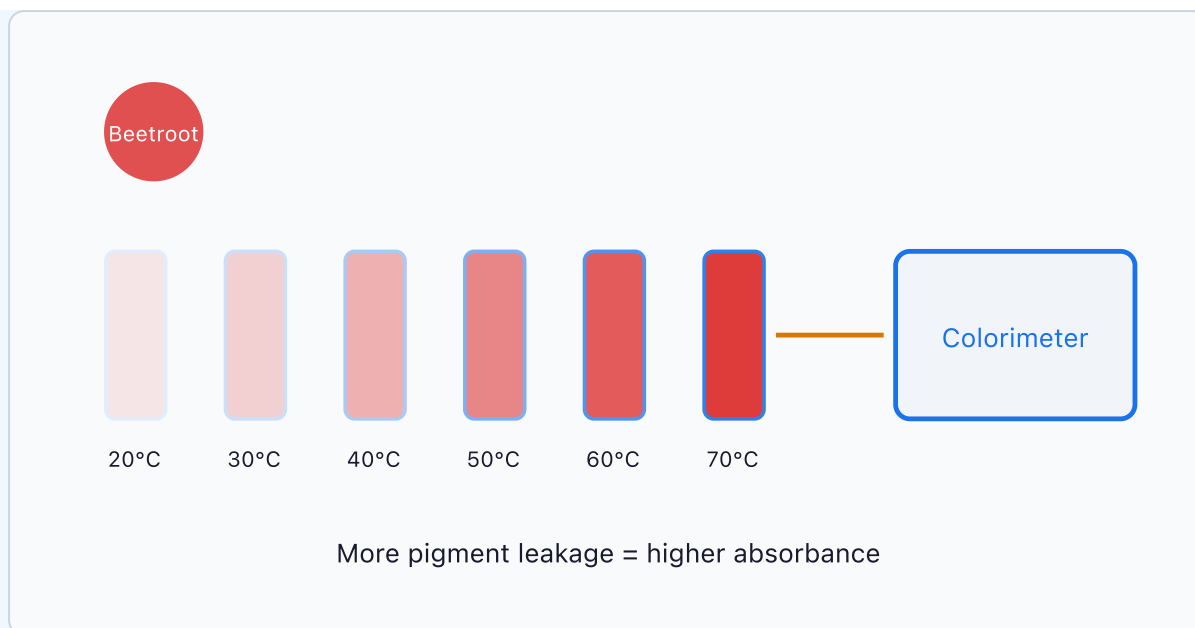
Sources of Error and Improvements

Error	Effect	Improvement
Subjective judgement of the colourless end-point	Inconsistent results between experimenters; difficult to judge exact moment of decolourisation	Use a colorimeter with a red filter (~600 nm, the light that blue DCPIP absorbs most strongly) to measure absorbance objectively. End-point defined as a specific absorbance value

Juice added too quickly (overshooting end-point)	More juice recorded than actually needed	Add drop by drop from a burette or graduated pipette; swirl after each addition; approach end-point slowly
Vitamin C in juice degrades in light and air (oxidation)	Lower apparent vitamin C concentration than the actual original content	Test immediately after opening/preparing; keep samples covered in foil; minimise exposure to air
Pulp or suspended particles in juice	Colour change obscured; difficult to see end-point	Filter juice through muslin or fine mesh before testing
Temperature variation between samples	Rate of reaction between vitamin C and DCPIP varies	Keep all solutions at the same temperature (room temperature)

Exam Tip: Vitamin C (ascorbic acid) is a **reducing agent** that reduces (decolourises) the blue dye DCPIP. More vitamin C means *less* volume of juice needed. Always state the end-point clearly: blue → colourless. If asked about a calibration curve, explain that you plot volume of known ascorbic acid (x) against concentration (y) and use interpolation to find the concentration of unknown samples. Common extension: why does boiled juice need more volume? Because vitamin C is destroyed by heat.

CP3: Membrane Permeability (Temperature or Alcohol Concentration)



Background Theory

Beetroot cells contain the red-purple pigment **betacyanin** (a type of betalain) stored in the **vacuole**, enclosed within the tonoplast membrane. Under normal conditions, the intact cell surface membrane and tonoplast are selectively permeable and retain the pigment inside the cell. According to the **fluid mosaic model**, membranes consist of a phospholipid bilayer with embedded proteins. At elevated temperatures: (1) the phospholipid bilayer becomes more fluid as phospholipids gain kinetic energy and move apart, (2) membrane channel and carrier proteins **denature** (their tertiary structure unfolds), creating gaps in the membrane, (3) the membrane loses its selective permeability. This allows betacyanin to leak out of the cell, colouring the surrounding solution. The degree of colour (measured by colorimeter absorbance) is a quantitative measure of membrane damage.

Apparatus

- Fresh beetroot, cork borer (e.g. size 4), scalpel, white tile, ruler
- Test tubes and rack, distilled water (measured volumes)
- Water baths set at 20, 30, 40, 50, 60, 70°C (thermostatically controlled)
- Colorimeter (set to blue-green filter, ~530 nm for betacyanin) and cuvettes, thermometer

Method

1. Use a cork borer to cut **uniform beetroot cylinders**. Trim to identical length (e.g. 1 cm) using a ruler and scalpel on a white tile. This ensures equal surface area for each replicate.
2. Rinse the cylinders **thoroughly** in cold distilled water until the water runs clear. This removes pigment released during cutting from damaged cells, which would give falsely high readings.
3. Place one cylinder in each test tube containing **5 cm³** distilled water (measure accurately with a measuring cylinder).
4. Place test tubes in water baths at the designated temperatures. Leave for **exactly 20 minutes** (use a stopwatch; stagger starts if necessary).
5. Remove the beetroot cylinder from each tube. Gently shake the solution to ensure even distribution of any leaked pigment.
6. Measure the **absorbance** (or percentage transmission) of each solution using a colorimeter with a blue-green filter (~530 nm). Zero the colorimeter using distilled water as a blank.
7. Repeat the entire procedure at least **3 times per temperature**. Calculate mean absorbance.

Alcohol version: the specification allows either temperature or alcohol concentration to be the independent variable. To test alcohol, place beetroot cylinders in a range of ethanol concentrations (e.g. 0–50% by volume, made up with distilled water) at a single fixed temperature for the same time, then measure the absorbance of each solution in the same way.

Key Variables

Variable	Detail
Independent	Temperature of water bath (20–70°C)
Dependent	Absorbance of the solution (arbitrary units) — a measure of betacyanin leakage
Control variables	Size and surface area of beetroot cylinders, volume of distilled water, time in water bath, same beetroot source, same colorimeter wavelength, thorough initial rinsing

Safety

- Sharp cutting tools — cut on a white tile, always cutting away from the body. Use a cork borer guard if available.

- Hot water baths — use tongs or forceps to handle test tubes in hot water. Take care to avoid burns.
- Beetroot juice stains skin and clothing — wear a lab coat and gloves.

Analysis

- Plot **mean absorbance (y-axis)** against **temperature (x-axis)**.
- Expected result: low, relatively constant absorbance from 20–40°C (membranes mostly intact), followed by a **sharp increase above ~40–50°C** as membrane proteins denature and phospholipid fluidity increases beyond the point of membrane integrity.
- Explain the sigmoid-like curve in terms of the fluid mosaic model: at low temperatures, the membrane remains selectively permeable; above ~40°C, protein denaturation and increased phospholipid movement cause progressive loss of membrane integrity; by 70°C, membranes are essentially destroyed and maximum pigment leakage has occurred.
- If alcohol concentration is the independent variable, absorbance rises as ethanol concentration increases, because ethanol dissolves in the phospholipid bilayer and disrupts it (and can denature membrane proteins).

Sources of Error and Improvements

Error	Effect	Improvement
Unequal cylinder sizes (different surface area:volume ratios)	Different amounts of pigment available for leakage	Use the same cork borer throughout; cut cylinders to the same length with a ruler; weigh cylinders for consistency
Insufficient initial rinsing	Surface pigment from cut cells inflates absorbance readings at all temperatures, including controls	Rinse thoroughly in running distilled water until the rinse water is completely clear
Timing inconsistency between tubes	Some tubes exposed for longer — confounding variable	Use a precise stopwatch; stagger tube placement at 1–2 minute intervals; record exact times

Water bath temperature fluctuations	Actual temperature differs from set temperature	Use thermostatically controlled water baths; check temperature with a thermometer before and after
Air bubbles or debris in cuvette	Inaccurate absorbance readings	Wipe cuvettes with lint-free tissue; check for bubbles; use the same orientation

Exam Tip: Always explain the increase in permeability in terms of the **fluid mosaic model**: (1) membrane proteins (channel proteins, carrier proteins) denature — their tertiary structure changes, creating gaps, (2) the phospholipid bilayer becomes more fluid as increased kinetic energy disrupts hydrophobic interactions between fatty acid tails, and (3) at very high temperatures, the bilayer may begin to break apart entirely. This allows the red pigment betacyanin to leak out by diffusion from the vacuole. A common exam mistake is saying "the membrane melts" — this is incorrect. The key process is protein **denaturation** and increased **membrane fluidity**.

CP4: Effect of Enzyme and Substrate Concentration on Initial Rate

Background Theory

This practical investigates how **enzyme concentration** and **substrate concentration** affect the initial rate of an enzyme-catalysed reaction, typically using **catalase** (which decomposes hydrogen peroxide into water and oxygen: $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$). According to the **lock-and-key** and **induced-fit** models, at low substrate concentrations, enzyme active sites are not all occupied, so increasing substrate concentration increases the rate proportionally (first-order kinetics). At high concentrations, **all active sites are saturated** (occupied by substrate), and the rate reaches a maximum (V_{max}) that cannot be exceeded without adding more enzyme. This produces the characteristic hyperbolic curve of rate vs substrate concentration.

Apparatus

- Catalase source: uniform potato discs (cut with cork borer) or measured volumes of liver homogenate
- Hydrogen peroxide at various concentrations: 0.5%, 1%, 2%, 4%, 8% (vol/vol), prepared by dilution from stock
- Conical flask, bung with gas delivery tube, gas syringe (or inverted measuring cylinder over water trough)
- Water bath at constant 25°C, stopwatch, measuring cylinder, balance

Method

1. Prepare potato discs of uniform size using a cork borer, then cut to the same mass. Alternatively, use a fixed volume of liver homogenate. Place the enzyme source in a conical flask.
2. Pre-warm the H_2O_2 solution in the water bath for 5 minutes to ensure it is at 25°C.
3. Add the H_2O_2 to the flask and **immediately** fit the bung and start the stopwatch. Speed is critical to capture the initial rate.
4. Record the volume of O_2 collected in the gas syringe every 30 seconds for 5 minutes.
5. Repeat with each H_2O_2 concentration. Perform **3 repeats** per concentration and calculate the mean.
6. To investigate **enzyme concentration**, keep the H_2O_2 concentration and volume fixed and vary the amount of enzyme instead (e.g. number of identical potato discs, or a range of dilutions of catalase or liver homogenate).
7. To determine the **initial rate** of reaction: draw a tangent to the curve at time = 0 on the volume vs time graph and calculate its gradient.

Safety

- Hydrogen peroxide is an **irritant** and can bleach skin and clothing at higher concentrations — wear eye protection and gloves. Use concentrations $\leq 8\%$.
- O_2 supports combustion — no naked flames in the vicinity.
- Take care when fitting the bung tightly — pressure can build up.

Analysis

- Plot **volume of O₂ (y-axis) against time (x-axis)** for each concentration. Calculate initial rate from the gradient of the tangent at $t = 0$.
- Plot **initial rate (y-axis) against substrate concentration (x-axis)**. Expected: a curve that rises steeply at first (active sites progressively filled), then plateaus at $V_{\max}$ (all active sites occupied — enzyme saturation).
- Plot **initial rate against enzyme concentration**. With substrate in excess, initial rate is directly proportional to enzyme concentration (more active sites available). The line levels off only if substrate becomes limiting.
- K_m (Michaelis constant) is the substrate concentration at which the rate is half $V_{\max}$. A low K_m indicates high enzyme-substrate affinity.

Sources of Error and Improvements

Error	Effect	Improvement
Gas escapes before the bung is fitted	Volume of O ₂ recorded is lower than actual (especially at high concentrations where the reaction is fastest)	Practise fitting the bung rapidly; use a side-arm flask with a rubber septum and inject H ₂ O ₂ through the septum to start the reaction without opening the flask
Variable amount of enzyme between replicates	Unreliable rate comparisons	Use the same mass of potato discs (weigh on a balance) or the same volume of homogenate measured with a syringe
Temperature variation during the experiment	Affects enzyme activity independently of substrate concentration (confounding variable)	Use a thermostatically controlled water bath; monitor temperature throughout
Using "total volume at 5 minutes"	Substrate depletion means later readings do not reflect	Calculate initial rate from the tangent to the curve at $t = 0$, which reflects the

instead of initial rate	the true rate at that concentration	rate when substrate concentration is at its starting value
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Exam Tip: At low substrate concentrations, rate increases proportionally because there are many unoccupied active sites (first-order kinetics). At high concentrations, all active sites are occupied (enzyme-substrate complexes forming as fast as they dissociate), so rate plateaus at $V_{\max}$ (zero-order kinetics). To increase $V_{\max}$, you must increase the enzyme concentration. Always use **initial rate** for valid comparisons — explain why (substrate concentration decreases over time as it is consumed, so later measurements do not reflect the starting concentration).

CP5: Root Tip Squash to Observe the Stages of Mitosis

Background Theory

In plants, cell division by **mitosis** is restricted to **meristems**. The apical meristem just behind a root tip (the first 1–2 mm) contains many actively dividing cells, so a squash of this region shows cells at every stage of the cell cycle. Most cells are in **interphase** (DNA replication and growth), with smaller numbers in **prophase** (chromosomes condense and become visible), **metaphase** (chromosomes line up on the equator, attached to spindle fibres by their centromeres), **anaphase** (sister chromatids pulled to opposite poles) and **telophase** (nuclear envelopes re-form; a cell plate begins to divide the cytoplasm). Warm **hydrochloric acid** kills the cells, stops division at whatever stage each cell has reached, and breaks down the pectin in the middle lamella so the cells can be spread into a single layer. A DNA stain such as **toluidine blue** or acetic orcein makes the chromosomes visible.

Apparatus

- Garlic cloves or onion bulbs grown over water for a few days until roots are 1–2 cm long
- 1 mol dm⁻³ hydrochloric acid in a water bath at about 55–60°C, watch glass or small tube
- Toluidine blue (or acetic orcein) stain, distilled water, dropping pipettes

- Microscope slides, coverslips, scalpel, forceps, mounted needle, filter paper
- Light microscope with $\times 10$ and $\times 40$ objectives

Method

1. Cut off the terminal 5 mm of a few root tips and place them in warm 1 mol dm^{-3} HCl for about 5 minutes.
2. Remove the tips with forceps and rinse them in cold distilled water, then blot dry.
3. Place one tip on a clean slide and cut away everything except the **1–2 mm at the very tip** (the meristem).
4. Add one or two drops of stain and break the tissue up with a mounted needle. Leave for about 2 minutes.
5. Lower a coverslip on top, cover with filter paper, and press down **firmly and vertically** with your thumb to squash the tissue into a single layer of cells. Do not twist or slide the coverslip.
6. Examine under low power to find the region of small, square, densely packed cells, then switch to high power.
7. Count the total number of cells in the field of view and the number in each stage of mitosis. Repeat for several fields of view.

Key Variables

Variable	Detail
What is measured	Number of cells in each stage; total number of cells in the field of view
Control variables	Time and temperature in HCl, staining time, region of root sampled (meristem only), same magnification for all counts
Possible extension	Compare mitotic index at different distances from the tip, or in roots grown at different temperatures

Safety

- Hydrochloric acid is an **irritant** — wear eye protection and wash any splashes off skin immediately.
- Stains mark skin and clothing — wear gloves and a lab coat.
- Take care with the scalpel; press the coverslip gently enough not to break it, as broken glass can cut.

Analysis

- Mitotic index** = number of cells undergoing mitosis ÷ total number of cells counted (often expressed as a percentage).
- The proportion of cells in a stage is proportional to the time that stage takes. If the whole cell cycle length is known, time in a stage = (cells in that stage ÷ total cells) × cell cycle time.
- Draw and label a few cells showing each stage. Chromosomes should be drawn as single continuous lines, with no shading.
- A high mitotic index shows rapid growth; the index falls with distance from the tip as cells stop dividing and begin to elongate and differentiate.

Sources of Error and Improvements

Error	Effect	Improvement
Cells overlapping (squash too thick)	Chromosomes hard to see; cells counted twice or missed	Use only the terminal 1–2 mm; macerate well; squash firmly and vertically
Tissue taken from outside the meristem	Few or no dividing cells; mitotic index underestimated	Cut precisely at the very tip; check under low power for small, square cells
Uncertainty in identifying stages (e.g. early prophase vs interphase)	Counts for different stages are inconsistent	Agree clear criteria before counting; count many cells across several fields of view
Counting cells at the edge of the field twice	Inaccurate totals	Only count cells cut by two chosen edges of the field (e.g. top and left)

Exam Tip: Learn what each step does: the **acid** hydrolyses the middle lamella (separating cells) and fixes the cells; the **stain** binds to DNA so chromosomes show up; **squashing** gives a single layer of cells so light can pass through. Only the root tip is used because that is where the meristem is. You may be asked to calculate a mitotic index, or to work out the time spent in a stage from the proportion of cells in it.

CP6: Identifying Sclerenchyma, Xylem and Phloem in Plant Stems

Background Theory

Plant stems contain **vascular bundles**. In a young dicotyledonous stem (e.g. sunflower, or the petiole of celery) the bundles form a ring near the outside. Each bundle contains three tissues that you must be able to identify:

- **Xylem vessels** — towards the centre of the bundle. Dead, hollow, elongated cells joined end to end with no end walls. Their walls are thickened with **lignin**, often in spirals or rings. They transport water and mineral ions and give support.
- **Phloem sieve tubes** — towards the outside of the bundle. Living cells with little cytoplasm and no nucleus, joined by perforated **sieve plates**, each with a small **companion cell**. Walls are cellulose (not lignified). They carry out translocation of organic solutes such as sucrose.
- **Sclerenchyma fibres** — a cap outside the phloem. Long, narrow, dead cells with tapering ends and very thick lignified walls. They provide support and are the source of useful plant fibres.

A stain that shows up lignin lets you tell these tissues apart: lignified walls (xylem, sclerenchyma) stain differently from unlignified walls (phloem, parenchyma).

Apparatus

- Celery petiole or young sunflower stem, new single-edged razor blade or sharp scalpel, white tile

- Toluidine blue O stain (or phloroglucinol with acid, used by the teacher), distilled water, watch glass, fine paintbrush
- Microscope slides, coverslips, light microscope, eyepiece graticule
- Pencil and plain paper for drawing

Method

1. Cut a short length of stem. Holding it upright on a tile, cut several **very thin transverse sections** (as thin as possible, even if incomplete) with the razor blade. Keep sections in water so they do not dry out.
2. Choose the thinnest sections and transfer them with the paintbrush into toluidine blue for about 1 minute, then rinse in water.
3. Mount a section in a drop of water on a slide and add a coverslip.
4. Under low power, locate the vascular bundles and draw a **low-power plan** showing the position of the tissues (outlines of tissue areas only, no individual cells).
5. Under high power, identify xylem vessels, phloem sieve tubes and sclerenchyma fibres by their size, wall thickness and staining colour.
6. Optionally, cut a **longitudinal section** through a bundle, or macerate a small piece of stem, to see the spiral thickening of xylem and the long tapering shape of sclerenchyma fibres.

Key Observations (Toluidine Blue)

Tissue	Position in bundle	Appearance
Xylem vessels	Inner side (towards centre of stem)	Large, empty, rounded cells; lignified walls stain blue-green
Phloem	Outer side of the xylem	Small cells with thin walls; stain purple (no lignin)
Sclerenchyma fibres	Cap on the outside of the phloem	Small cells with very thick walls and narrow lumens; stain blue-green

Safety

- Razor blades are extremely sharp — cut away from your fingers, onto a tile, and use a blade with a guarded back edge.
- Toluidine blue stains skin — wear gloves. Phloroglucinol with acid is corrosive and should only be handled with eye protection, ideally by the teacher.

Analysis

- Produce a labelled plan showing epidermis, cortex, vascular bundles (with xylem, phloem and sclerenchyma), and pith.
- Use an eyepiece graticule calibrated against a stage micrometer to measure the diameter of a xylem vessel and a sclerenchyma fibre.
- Relate structure to function: hollow, lignified xylem for water transport and support; sieve plates and companion cells for translocation; thick-walled sclerenchyma for strength.

Sources of Error and Improvements

Error	Effect	Improvement
Sections too thick	Cells overlap; light cannot pass through; tissues hard to identify	Use a new, sharp blade; cut many sections and choose the thinnest; a thin wedge is fine
Section allowed to dry out	Cells shrink and distort; air bubbles	Keep sections in water at all times; mount in water
Over-staining	All tissues look dark; colour differences lost	Stain for a short, fixed time and rinse before mounting

Exam Tip: Be able to **identify the three tissues from a photomicrograph or drawing** and state where each is found in the stem. Xylem and sclerenchyma are both lignified and dead, but xylem vessels are wide tubes with no end walls, while sclerenchyma fibres are narrow, with tapering ends and almost no lumen. Phloem is the only one of the three that is living. Low-power plans show tissue areas only — do not draw individual cells.

CP7: Plant Mineral Deficiencies

Background Theory

Plants absorb inorganic ions from the soil through their roots. Without them, plants show characteristic **deficiency symptoms**:

- **Nitrate ions** — needed to make amino acids (and so proteins), nucleotides (DNA, RNA, ATP) and chlorophyll. Deficiency causes stunted growth and yellowing (**chlorosis**), starting in older leaves.
- **Magnesium ions** — part of the **chlorophyll** molecule. Deficiency causes chlorosis, especially between the veins of older leaves, and reduced photosynthesis.
- **Calcium ions** — form **calcium pectate** in the middle lamella, which holds neighbouring cell walls together. Deficiency affects the growing points first: new leaves and roots are small and distorted.

The practical grows plants in culture solutions that are identical except that one ion is missing, so any difference in growth can be linked to that ion.

Apparatus

- Duckweed (*Lemna*) plants, or seedlings such as radish or wheat
- Culture solutions: complete, lacking nitrate, lacking magnesium, lacking calcium; plus distilled water
- Sterile Petri dishes, tubes or small bottles with lids, labels
- Aluminium foil or black paper (to stop algae growing in tubes), light bank, hand lens, balance

Method

1. Fill labelled containers with equal volumes of each culture solution.
2. Add the same number of duckweed plants (e.g. 3 plants with 3 fronds each) to each container, choosing healthy plants of similar size and colour.
3. Cover the containers (to reduce evaporation and contamination) and place them all in the same position under the same light source and at the same temperature.

4. Count the number of fronds in each container at the start and every few days for 2–3 weeks. Record frond colour and any unusual appearance.
5. Set up at least 3 replicates of each solution.
6. At the end, blot and weigh the plants from each container (or dry them to find dry mass).

Key Variables

Variable	Detail
Independent	Culture solution (complete, or lacking one named ion)
Dependent	Number of fronds; colour of fronds; final mass
Control variables	Volume of solution, starting number and size of plants, light intensity and duration, temperature, time

Safety

- Culture solutions are low hazard, but wash hands after handling them and the plants.
- Keep electrical light banks away from water.

Analysis

- Plot the number of fronds against time for each solution, or calculate the mean percentage increase in fronds.
- The complete solution should give the most growth, with green fronds. Solutions lacking nitrate or magnesium give pale, yellowish fronds and less growth; distilled water gives the least growth.
- Explain each result in terms of what the missing ion is used for (proteins and nucleic acids, chlorophyll, middle lamella).
- A statistical test (e.g. a t-test) can compare the mean final frond number in the complete solution with that in a deficient solution.

Sources of Error and Improvements

Error	Effect	Improvement
Algae or microorganisms growing in the solution	They compete for ions and light; results unreliable	Use sterile containers and solutions; cover containers; wrap tubes in foil to keep roots dark
Plants start with different ion reserves	Deficiency symptoms delayed or reduced	Use plants of the same age and size from the same stock; run the experiment for long enough
Colour judged by eye	Subjective and hard to compare	Use a colour chart, or extract chlorophyll and measure absorbance with a colorimeter
Evaporation changes ion concentrations	Solutions become more concentrated over time	Cover containers; top up with distilled water to a marked level

Exam Tip: Link each deficiency to the ion's job: **nitrate** for amino acids, proteins, DNA and chlorophyll; **magnesium** for chlorophyll; **calcium** for calcium pectate in the middle lamella. Explain why the complete culture solution is needed as a control: it shows how the plants grow when no ion is missing, so any reduced growth can be put down to the missing ion.

CP8: Tensile Strength of Plant Fibres

Background Theory

Plant cell walls contain **cellulose microfibrils** — long, unbranched chains of beta-glucose linked by 1,4-glycosidic bonds, held together by hydrogen bonds into bundles. These microfibrils provide exceptional tensile strength (resistance to stretching/breaking). **Sclerenchyma fibres** have additional secondary cell wall thickening and are often **lignified**, making them extremely strong. This practical measures the tensile strength of plant fibres by progressively loading them

until they break, and relates the breaking force to the cross-sectional area of the fibre. Tensile stress = force / cross-sectional area.

Apparatus

- Plant fibres (e.g. celery petiole strings, nettle bast fibres, sclerenchyma strands)
- Clamp stand, boss, clamp (or a purpose-built fibre-testing rig)
- Masses (10 g, 20 g, 50 g increments), hook or S-clip, container (to catch fallen masses)
- Ruler, micrometer (eyepiece graticule and stage micrometer) or digital callipers for measuring fibre diameter

Method

1. Carefully extract a single fibre from the plant using forceps and a mounted needle. Avoid damaging the fibre during extraction.
2. Measure the **diameter** of the fibre at several points along its length using a micrometer or calibrated microscope eyepiece graticule. Calculate the mean diameter and the radius (r). Calculate cross-sectional area: $A = \pi r^2$.
3. Clamp one end of the fibre securely. Attach a hook to the lower end.
4. Add masses one at a time (e.g. 10 g increments). Wait ~10 seconds after each addition for the fibre to stabilise.
5. Record the mass at which the fibre **breaks**.
6. Repeat for at least **10 fibres** per plant species to obtain a reliable mean. Calculate mean breaking mass.
7. Convert mass to force: $F = m \times g$ (where $g = 9.81 \text{ N/kg}$).
8. Calculate **tensile stress** = F / A (in Pa or MPa).

Safety

- Place a cushion, tray, or container below the hanging masses to catch them if they fall when the fibre breaks.
- Wear eye protection in case the fibre snaps suddenly and recoils.
- Take care when extracting fibres (sharp tools may be needed).

Analysis

- **Force at break** = mass (kg) x 9.81 (N/kg). Convert g to kg first.
- **Cross-sectional area** = πr^2 (assuming a circular cross-section).
- **Tensile stress** = force / area. Report in MPa (megapascals). Higher stress = stronger fibre.
- Compare fibres from different species, or from different parts of the same plant (e.g. stem vs petiole).
- Relate strength to cell wall composition: cellulose microfibril density, lignification, secondary wall thickness.

Sources of Error and Improvements

Error	Effect	Improvement
Fibre damaged during extraction	Weakened structure; breaks at lower force than its true tensile strength	Extract carefully using fine forceps and a mounted needle; avoid bending, kinking, or stretching
Variable fibre diameter along its length	Cross-sectional area calculation inaccurate; stress may be over- or underestimated	Measure diameter at multiple points (e.g. 3–5 positions); calculate mean diameter; acknowledge this as a source of uncertainty
Fibre slips out of clamp instead of breaking	Records a lower apparent breaking force; not a true measure of tensile strength	Wrap the fibre securely around the clamp; use rubber pads to grip; ensure the clamp is tight
Small sample size	Mean may not be representative; high uncertainty	Test at least 10 fibres per type; report standard deviation and range
Fibre cross-section is not circular	$A = \pi r^2$ assumes a circle; actual area may differ	Examine cross-sections under a microscope; use image analysis software for more accurate area measurement

Exam Tip: Tensile strength is primarily due to **cellulose microfibrils** in the cell wall, which run in parallel and are cross-linked by hydrogen bonds. **Sclerenchyma fibres** are particularly strong because they have thick secondary walls with additional cellulose layers and **lignin** deposits (which also makes them waterproof). Compare with collagen fibres in animals, which provide tensile strength in tendons.

CP9: Antimicrobial Properties of Plants (Aseptic Technique)

Background Theory

Many plants make chemicals that kill or inhibit microorganisms, as a defence against infection. Examples include garlic, mint, ginger, basil and tea tree. Historically these plants were used as medicines, and plant extracts are still screened for new antimicrobial drugs. In this practical, an extract is soaked onto a paper disc and placed on a **lawn of bacteria** on agar. Antimicrobial substances diffuse out of the disc; where they are concentrated enough to stop bacterial growth, a clear **zone of inhibition** forms. The bigger the clear zone, the more effective the extract (or the better it diffuses through agar). Because bacteria are handled, **aseptic technique** is essential: it keeps unwanted microorganisms out of the culture and keeps the bacteria away from people and the lab.

Apparatus

- Fresh plant material (e.g. garlic, mint, basil), pestle and mortar, sand, ethanol (as solvent), filter or muslin
- Sterile filter-paper discs, sterile forceps, Petri dishes of nutrient agar
- Broth culture of a non-pathogenic bacterium (e.g. *Bacillus subtilis* or *E. coli* K-12)
- Sterile pipette and spreader, Bunsen burner, disinfectant, adhesive tape, marker pen, ruler, incubator at 25°C

Method

1. Grind a known mass of each plant with a fixed volume of ethanol and a little sand. Filter to obtain the extract.
2. Wipe the bench with disinfectant and work close to a lit Bunsen burner (on a blue flame).
3. Using a sterile pipette, transfer a small volume of bacterial culture to an agar plate, lifting the lid only slightly. Spread it evenly over the surface with a sterile spreader to make a lawn.
4. Using sterile forceps, dip a paper disc in one extract. Let the ethanol evaporate for a few minutes (away from the flame) so the solvent does not affect the bacteria.
5. Place the discs on the agar, well spaced. Include a **control disc** soaked in ethanol only and allowed to evaporate.
6. Tape the lid on with two short strips (do not seal all round), label the base, and incubate **upside down at 25°C** for 48 hours.
7. Without opening the plate, measure the diameter of each clear zone through the base and calculate its area.

Key Variables

Variable	Detail
Independent	Type of plant extract
Dependent	Area of the zone of inhibition (mm ²)
Control variables	Mass of plant and volume of solvent used for each extract, disc size, soaking and drying time, type and volume of bacterial culture, incubation temperature and time

Safety

- **Ethanol is highly flammable** — keep it and the soaked discs well away from the Bunsen flame.
- Treat all cultures as potentially harmful: use aseptic technique, incubate at no more than 25°C, and never open plates after incubation.
- Wash hands before and after; disinfect the bench; autoclave all plates and cultures before disposal.

Analysis

- Area of clear zone = πr^2 , where r = diameter $\div$ 2. Subtract the area of the disc if needed.
- Calculate the mean area for each extract from several discs or plates; compare using a statistical test.
- The control disc should have no clear zone, showing that the solvent is not responsible for any inhibition.
- Remember that zone size depends on both how strongly the chemical acts and how well it diffuses through the agar.

Sources of Error and Improvements

Error	Effect	Improvement
Different concentrations of extract	Unfair comparison between plants	Use the same mass of plant per volume of solvent; grind for the same time
Solvent not fully evaporated	Ethanol itself inhibits bacteria, enlarging zones	Allow a fixed drying time; always include a solvent-only control disc
Contaminating microorganisms	Unexpected colonies make zones hard to measure	Strict aseptic technique; sterile equipment; lids lifted as little as possible
Zones not circular	Area calculation less accurate	Measure two diameters at right angles and use the mean

Exam Tip: Questions often focus on **aseptic technique**: disinfect the bench, work near a Bunsen flame, flame the necks of bottles and sterilise loops, lift lids as little as possible, tape (do not fully seal) the lid, incubate at 25°C and upside down. Give a reason for each. Also explain why the solvent-only control disc is needed.

CP10: Ecology of a Habitat (Quadrats, Transects and Abiotic Factors)

Background Theory

Ecologists rarely count every organism in a habitat. Instead, they use **sampling techniques** to estimate population size and species diversity. The core techniques for this practical are **quadrats** and **transects** for plants and slow-moving or sessile organisms, together with measurement of **abiotic factors**. **Mark-release-recapture (Lincoln index)** is a useful extra for mobile animals, but it is not named in the 9BN0 specification. **Random sampling** is essential to avoid bias — use random number tables or a random number generator to determine coordinates for quadrat placement. An **index of diversity** quantifies biodiversity in a habitat, accounting for both species richness (number of species) and evenness (relative abundance of each species).

Apparatus

- Quadrats (e.g. 0.25 m² or 1 m²), tape measures (for laying out a grid), random number table or calculator
- Pitfall traps, sweep nets, pooters, or pond nets (for mark-release-recapture of invertebrates)
- Non-toxic, non-harmful marker (waterproof paint pen, correction fluid, nail varnish — for marking animals)
- Identification keys, field guides, recording sheets, tally counter, calculator
- Abiotic measuring equipment: light meter, thermometer, soil moisture meter, pH probe or soil test kit, anemometer

Method

A. Quadrat Sampling (for plants and sessile organisms):

1. Lay out two tape measures at right angles along two edges of the study area to create a coordinate grid.
2. Generate random pairs of coordinates using a random number table.
3. Place the quadrat at each set of random coordinates.

4. For each quadrat: record the species present and either **count individuals** (for countable organisms), estimate **percentage cover** (for spreading plants like grasses, mosses), or use an **ACFOR scale** (Abundant, Common, Frequent, Occasional, Rare).
5. Use enough quadrats to give a representative sample (typically 10–30 depending on area size and habitat heterogeneity). The mean number of individuals per quadrat multiplied by the total number of quadrats that would fit in the area gives an estimate of total population.

B. Mark-Release-Recapture (Lincoln Index) (for mobile animals; optional extension, not named in the specification):

1. **Capture** a sample of individuals using an appropriate method (pitfall traps for ground beetles, sweep nets for meadow invertebrates). Count and record the number captured = **M**.
2. **Mark** each individual with a small, non-toxic, non-conspicuous mark (e.g. a dot of nail varnish on the wing case of a beetle). The mark must not affect survival, predation risk, or behaviour.
3. **Release** all marked individuals back into the habitat. Allow sufficient time for them to mix randomly with the unmarked population (at least 24 hours for most invertebrates).
4. **Recapture** a second sample from the same area. Count the total number in the second sample = **n**. Count the number of marked individuals in the second sample = **m**.
5. Estimate total population: $N = (M \times n) / m$ (Lincoln index formula).

C. Transects and Abiotic Factors (for changes in distribution across a habitat):

1. Where conditions change across a habitat (e.g. from woodland into open grassland, or up a sand dune or rocky shore), lay a tape measure along the gradient as a **transect**.
2. For a **belt transect**, place quadrats at regular intervals (e.g. every 2 m) along the line and record the abundance of each species (count, percentage cover or ACFOR). A **line transect** records only species touching the tape at each point.
3. At each sampling point, measure the **abiotic factors** likely to affect distribution, e.g. light intensity, temperature, soil moisture, soil pH, wind speed. Take all readings at about the same time of day.
4. Repeat the transect at least three times in parallel lines and calculate means.
5. Plot a kite diagram of species abundance along the transect, with the abiotic readings below it, and use Spearman's rank correlation to test for a relationship between an abiotic factor and a species' abundance.

Safety

- Complete a fieldwork risk assessment (weather, terrain, slippery surfaces, water bodies, stinging nettles, ticks).
- Handle all organisms gently and return them to their habitat promptly.
- Wash hands thoroughly after handling soil, water, or organisms.
- Wear appropriate clothing and footwear for the terrain.

Analysis

Lincoln Index: $N = (M \times n) / m$

Assumptions (must be stated for full marks):

- The marks do not affect survival or make organisms more visible to predators
- The marks do not rub off or fade between sampling periods
- The population is **closed** (no immigration, emigration, births, or deaths between samples)
- Sufficient time has elapsed for marked individuals to mix randomly with unmarked individuals
- Sampling methods are the same for both capture events (equal probability of capture)

Index of Diversity:

$$D = N(N - 1) \div \sum n(n - 1)$$

Where N = total number of organisms of all species, n = number of organisms of each species.

D = 1 for a habitat with a single species; the higher the value of D, the greater the biodiversity.

Sources of Error and Improvements

Error	Effect	Improvement
Marks wear off or are lost between sampling events	Fewer marked individuals found in recapture; m decreases; N is overestimated	Use permanent, waterproof markers; check mark durability beforehand

Marked animals more conspicuous to predators	Marked individuals are preferentially eaten; m decreases; N is overestimated	Use small, inconspicuous marks in locations not visible to predators
Immigration or emigration between sampling events	Population not closed; estimate becomes inaccurate	Sample within a short time frame; use a clearly bounded habitat; state the assumption
Non-random quadrat placement (e.g. avoiding muddy areas)	Biased species counts; unrepresentative of the true population	Use a random number table or random number generator for coordinates; accept whatever falls in the quadrat
Insufficient number of quadrats	Unrepresentative sample; high variance in estimates	Use at least 10–30 quadrats; increase number until the running mean stabilises

Exam Tip: If you use mark-release-recapture, the Lincoln index formula is $N = (M \times n) / m$; state its assumptions. If marks affect survival, the formula gives an **overestimate** (fewer marked individuals survive to be recaptured, so m is low, inflating N). Index of diversity: $D = N(N - 1) \div \sum n(n - 1)$. A habitat with many species in roughly equal numbers has a high D . A habitat dominated by one species has a low D , even if many species are present. You must be able to calculate D from given data, and to explain how transects and abiotic measurements show why a species is found where it is.

CP11: Photosynthesis Using Isolated Chloroplasts (the Hill Reaction)

Background Theory

In the **light-dependent reactions**, light energy excites electrons in chlorophyll. The electrons pass along the electron transport chain in the thylakoid membranes and are normally accepted

by **NADP**, which is reduced. Water is split (photolysis) to replace the lost electrons, releasing oxygen. In the **Hill reaction**, isolated chloroplasts are given an artificial electron acceptor, the blue dye **DCPIP**. When DCPIP accepts electrons it is reduced and turns **colourless**. The rate at which the blue colour disappears is therefore a measure of the rate of the light-dependent reactions.

Apparatus

- Fresh spinach or lettuce leaves, scissors, ice bath
- Ice-cold **isolation medium** (buffer solution containing sucrose or sorbitol)
- Chilled blender or pestle and mortar, muslin or filter, centrifuge and tubes (kept cold)
- DCPIP solution, test tubes, pipettes or syringes, aluminium foil
- Bench lamp, ruler, colorimeter with a red filter (or a white tile to judge colour), stopwatch

Method

1. Cut the leaves into small pieces (removing the midribs) and blend or grind them in ice-cold isolation medium.
2. Filter through several layers of muslin into a cold beaker to remove cell debris.
3. Centrifuge the filtrate at low speed to remove any remaining debris, then centrifuge the liquid at a higher speed to form a pellet of chloroplasts.
4. Pour off the liquid and resuspend the pellet in a small volume of ice-cold isolation medium. Keep this chloroplast suspension on ice and in dim light.
5. Set up tubes containing equal volumes of chloroplast suspension and DCPIP. Place tubes at different distances from a lamp (or use different light intensities).
6. Record the time taken for each tube to lose its blue colour, or measure the absorbance with a colorimeter at regular intervals.
7. Set up **controls**: (a) chloroplasts and DCPIP wrapped in foil (dark); (b) isolation medium and DCPIP without chloroplasts, in the light.

Key Variables

Variable	Detail
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Independent	Light intensity (distance from lamp), or another factor such as temperature
Dependent	Time for DCPIP to decolourise, or change in absorbance per minute
Control variables	Volume of chloroplast suspension and DCPIP, temperature (use a heat shield), time since extraction

Safety

- DCPIP is harmful if swallowed and stains — wear eye protection and gloves.
- Take care with blender blades; balance centrifuge tubes before spinning and keep the lid closed until the rotor stops.
- Keep the lamp and other electrical items away from water.

Analysis

- Calculate rate = $1 \div \text{time to decolourise}$, or the rate of fall in absorbance.
- Plot rate against light intensity (light intensity is proportional to $1/d^2$). The rate rises with light intensity and then levels off when another factor becomes limiting.
- The dark tube stays blue, showing that light is needed. The tube without chloroplasts stays blue, showing that chloroplasts are needed.

Sources of Error and Improvements

Error	Effect	Improvement
Chloroplasts damaged during extraction	Lower activity; decolourisation slow or absent	Keep everything ice-cold; use an isotonic, buffered medium; work quickly
Chloroplast suspension ageing during the experiment	Later tubes react more slowly	Use the suspension quickly; test in a random order; keep it on ice between uses
End-point judged by eye	Subjective; inconsistent	Use a colorimeter, or compare against a colour standard

Heat from the lamp	Temperature changes as well as light intensity	Use an LED lamp or a heat shield; monitor temperature
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Exam Tip: Explain each feature of the isolation medium: **ice-cold** to slow enzymes that would damage the chloroplasts; **isotonic** (same water potential) so the chloroplasts do not burst or shrink by osmosis; **buffered** to keep the pH constant so proteins are not denatured. DCPIP replaces NADP as the electron acceptor — it goes from blue to colourless when **reduced**.

CP12: Effect of Temperature on the Initial Rate of an Enzyme Reaction (including Q_{10})

Background Theory

Temperature affects enzyme activity through two competing mechanisms: (1) **increased kinetic energy** at higher temperatures means more frequent collisions between enzyme and substrate molecules, and more collisions have sufficient activation energy to form the enzyme-substrate complex (rate increases), (2) above the **optimum temperature**, the thermal energy breaks the weak bonds (hydrogen bonds, ionic bonds, hydrophobic interactions) maintaining the enzyme's tertiary structure, causing the active site to change shape (**denaturation**). Denaturation is irreversible — the substrate can no longer fit into the active site (induced-fit model fails). The result is a bell-shaped curve of rate vs temperature.

Apparatus

- Enzyme: amylase solution (for starch hydrolysis) or catalase source (for H_2O_2 decomposition)
- Substrate: 1% starch solution (for amylase) or H_2O_2 (for catalase)
- Water baths at 20, 30, 40, 50, 60°C (thermostatically controlled)
- For amylase: spotting tile, iodine solution (in potassium iodide), dropping pipette, stopwatch
- For catalase: gas syringe, conical flask with bung and delivery tube, stopwatch

- Thermometer, test tubes, syringes for measuring volumes

Method (Using Amylase and Starch)

1. Place tubes of **amylase** and **starch** solution in separate test tubes in the same water bath. **Equilibrate for 5 minutes** to ensure both solutions reach the target temperature before mixing.
2. Meanwhile, prepare a spotting tile with drops of iodine solution in each well.
3. Mix the amylase and starch. Start the stopwatch immediately.
4. Every **30 seconds**, remove a small sample with a pipette and add it to a well of iodine on the spotting tile. If starch is still present, the iodine turns **blue-black**. When all starch has been digested, the iodine remains **yellow-brown** (end-point).
5. Record the **time taken** for the iodine to no longer change colour (starch fully hydrolysed).
6. Repeat at each temperature (3 repeats). Calculate **rate = 1/time**.

Safety

- Care with hot water baths (burns risk). Use test tube tongs.
- Iodine solution stains skin and clothing — wear gloves and lab coat.
- H_2O_2 is irritant — eye protection required.

Analysis

- Calculate rate = $1/\text{time}$ (s^{-1}) for each temperature.
- **Temperature coefficient:** $Q_{10} = \text{rate at } (T + 10)^\circ\text{C} \div \text{rate at } T^\circ\text{C}$. Calculate it for pairs of temperatures 10°C apart, below the optimum. For most enzyme-controlled reactions it is about 2 (the rate roughly doubles for each 10°C rise); it falls below 1 above the optimum as the enzyme denatures.
- Plot **rate (y-axis) against temperature (x-axis)**. Expect a bell-shaped curve with a clear optimum.
- Below the optimum: rate increases with temperature (more kinetic energy $\rightarrow$ more enzyme-substrate collisions $\rightarrow$ more ES complexes formed per unit time). $Q_{10} \approx 2$.
- At the optimum: maximum rate; all factors balanced.

- Above the optimum: rate decreases sharply as the enzyme **denatures** — weak bonds in the tertiary structure break, the active site changes shape irreversibly, and the substrate can no longer bind.

Sources of Error and Improvements

Error	Effect	Improvement
Not pre-equilibrating enzyme and substrate at the target temperature	The reaction starts at the wrong temperature; initial rate is unreliable	Always equilibrate both tubes separately for at least 5 minutes in the water bath before mixing
Delay between mixing and starting the stopwatch	Misses the initial period of the reaction; underestimates rate	Start the stopwatch at the exact moment of mixing; have the stopwatch ready
Subjective colour judgement for end-point (iodine method)	Inconsistent determination of when starch is fully digested	Use a colorimeter set to a red filter (~600 nm) for objective measurement of starch-iodine colour; define a numerical threshold
Water bath temperature fluctuates	Actual temperature differs from the set temperature	Use thermostatically controlled water baths; check temperature with a thermometer before and after

Exam Tip: Always equilibrate enzyme and substrate **separately** at the target temperature before mixing. This is critical — if you mix cold solutions and then heat them, the initial rate does not reflect the target temperature. Use **rate = $1/t$** for valid comparisons (strictly a mean rate used as an estimate of the initial rate; for a truer initial rate, follow the reaction continuously, e.g. catalase with a gas syringe, and take the gradient of the tangent at $t = 0$), and be able to calculate **$Q_{10} = \text{rate at } (T + 10)^{\circ}\text{C} \div \text{rate at } T^{\circ}\text{C}$** from a table or graph. The optimum temperature for human enzymes is typically $\sim 37^{\circ}\text{C}$; for thermophilic bacteria, it can be $>80^{\circ}\text{C}$. Denaturation is **permanent** (irreversible) — this is different from reversible

inhibition. Do not describe the enzyme as "killed" or "dead" — enzymes are not alive. Say "denatured."

CP13: Effect of Temperature on the Development of Organisms

Background Theory

Growth and development depend on enzyme-controlled reactions, so they are affected by temperature. As temperature rises towards the optimum, molecules have more kinetic energy, enzyme–substrate complexes form more often, and development speeds up. Above the optimum, enzymes begin to denature and development slows or fails. This practical links to the evidence for the effects of **climate change**: warmer temperatures can change growth rates, the timing of life-cycle events such as hatching, flowering and germination, and the distribution of species. Two convenient organisms are **brine shrimp** (*Artemia*), whose dormant eggs (cysts) hatch in salt water, and fast-germinating **seedlings** such as cress or mung beans.

Apparatus

- Brine shrimp cysts, artificial sea water (sea salt dissolved in dechlorinated water), balance, measuring cylinder
- Containers (e.g. beakers or tubes) with lids, water baths or incubators set at a range of temperatures (e.g. 10, 15, 20, 25, 30°C), thermometers
- Light source, dropping pipette, cavity tile or Petri dish, hand lens or low-power microscope
- Alternative: seeds, damp paper towels, ruler (for seedling growth)

Method (Brine Shrimp Hatch Rate)

1. Put equal volumes of artificial sea water into each container and let them reach the set temperatures.
2. Add the same mass of cysts (e.g. 0.05 g) to each container. Keep all containers in the same light conditions.

3. At regular intervals (e.g. every 12 or 24 hours), stir gently and remove a fixed volume with a pipette. Count the hatched larvae (nauplii) using a hand lens or microscope, then return them.
4. Continue for 2–3 days. Repeat each temperature at least three times.
5. **Seedling alternative:** germinate equal numbers of seeds at each temperature and measure the mean shoot length each day to find the growth rate.

Key Variables

Variable	Detail
Independent	Temperature
Dependent	Number of larvae hatched over time (or mean seedling growth per day)
Control variables	Mass of cysts, volume and salt concentration of the water, light intensity, sampling volume and times, source of cysts or seeds

Safety and Ethical Considerations

- Brine shrimp are invertebrates, but should still be treated with respect: use small numbers and dispose of them humanely at the end, as advised by your teacher.
- Keep electrical equipment away from water; wash hands after handling cultures and seeds.

Analysis

- Plot the mean number hatched against time for each temperature, or plot the mean number hatched after a fixed time against temperature.
- The rate of hatching (or growth) should increase with temperature up to an optimum, then fall at higher temperatures.
- A temperature coefficient can be calculated: $Q_{10} = \text{rate at } (T + 10)^{\circ}\text{C} \div \text{rate at } T^{\circ}\text{C}$.
- Discuss what the results suggest about how a warming climate might affect the development of organisms, and the limits of applying lab results to the wild.

Sources of Error and Improvements

Error	Effect	Improvement
Larvae not evenly spread when sampling	Counts vary between samples	Stir gently before each sample; take several samples and use the mean
Temperature fluctuates in containers	Actual temperature differs from the set value	Use thermostatically controlled water baths or incubators; check with a thermometer
Different numbers of cysts per container	Unfair comparison of numbers hatched	Weigh the cysts accurately; express results as a proportion of a sample hatched
Larvae hard to count as they move	Miscounts	Count in a gridded dish; photograph the sample and count from the image

Exam Tip: Explain the effect of temperature using **enzymes**: more kinetic energy means more successful collisions up to the optimum; above it, enzymes denature. Be ready to evaluate how far a lab investigation can support conclusions about climate change — in the wild, many other factors vary as well as temperature.

CP14: Gel Electrophoresis to Separate DNA Fragments

Background Theory

Gel electrophoresis separates DNA fragments according to their **length**. DNA is cut into fragments with **restriction enzymes**, which cut at specific base sequences. The fragments are placed in wells in an agarose gel and an electric current is passed through the gel. Because of their phosphate groups, DNA fragments are **negatively charged**, so they move towards the positive electrode (anode). The gel acts like a mesh: **shorter fragments move further** in a given time than longer ones. A **DNA ladder** (fragments of known length) run alongside lets the size of

unknown fragments be estimated. The technique is used in DNA profiling (for example in forensic science) and to compare DNA sequences between organisms.

Apparatus

- Agarose gel (about 0.8–1%) cast in a tray with a comb to form wells
- Electrophoresis tank, running buffer, power supply (low voltage)
- DNA samples already cut with restriction enzymes (e.g. lambda DNA digested with different enzymes), DNA ladder, loading dye
- Micropipette and tips, DNA stain safe for schools (e.g. methylene blue or Azure A), ruler, light box or white tile

Method

1. Place the gel in the tank with the wells at the **negative (cathode) end**. Cover the gel with running buffer.
2. Mix each DNA sample with loading dye. The dye makes the sample dense so it sinks into the well, and lets you see how far the run has progressed.
3. Using a micropipette, carefully load a fixed volume of each sample into a separate well without piercing the bottom of the well. Load the DNA ladder into one well.
4. Put the lid on, connect to the power supply and run the gel (e.g. 30–60 minutes, depending on voltage) until the dye front has moved most of the way along the gel.
5. Switch off and disconnect the power, then remove the gel and stain it to show the DNA bands.
6. Measure the distance each band has moved from its well.

Key Variables

Variable	Detail
What separates the fragments	Fragment length (all DNA has a similar charge-to-mass ratio)
Measured	Distance moved by each band

Control variables	Gel concentration, voltage and running time, buffer, volume of sample loaded
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Safety

- Electrical hazard: only switch on with the lid in place, and switch off and disconnect before touching the tank or gel.
- Wear gloves and eye protection when handling stains and buffer.
- Hot molten agarose can scald if the teacher is pouring gels.

Analysis

- Use the DNA ladder to draw a **calibration curve** of fragment length (usually on a log scale) against distance moved.
- Read off the length of each unknown fragment from its distance moved.
- Compare banding patterns: samples from the same source produce matching bands; in DNA profiling, a suspect's profile is compared with crime-scene DNA.

Sources of Error and Improvements

Error	Effect	Improvement
Well pierced during loading	Sample leaks under the gel; band faint or missing	Hold the tip just inside the top of the well; steady the pipette with both hands
Gel run for too long	Small fragments run off the end of the gel	Watch the dye front and stop before it reaches the end
Bands fuzzy or smeared	Hard to measure distances accurately	Use the correct gel concentration and voltage; load small, equal volumes
Measuring from different points	Inconsistent distances	Always measure from the well to the middle of the band

Exam Tip: Two points are regularly tested: DNA moves towards the **positive** electrode because of its negatively charged **phosphate groups**, and **shorter fragments travel**

further because they pass through the gel more easily. Be able to use a DNA ladder to estimate fragment sizes, and to interpret profiles, e.g. matching a suspect to crime-scene DNA or working out relationships from shared bands.

CP15: Effect of Different Antibiotics on Bacteria

Background Theory

This practical compares the effectiveness of different antibiotics against a bacterium. (Plant extracts are tested in a similar way in CP9.) When antimicrobial discs are placed on an agar plate seeded with bacteria, the agent **diffuses outward** through the agar, creating a concentration gradient. Where the concentration exceeds the minimum inhibitory concentration (MIC), bacteria are killed or inhibited, producing a clear zone called the **zone of inhibition**. A larger zone indicates a more effective antimicrobial agent (or one that diffuses more readily through the agar). This is the **Kirby-Bauer disc diffusion method**, widely used in clinical microbiology to guide antibiotic selection.

Apparatus

- Pre-poured nutrient agar plates, non-pathogenic bacterial broth culture (e.g. *E. coli* K-12 or *Micrococcus luteus*)
- Antibiotic multidiscs (or individual discs soaked in different antibiotics/plant extracts)
- Sterile glass spreader (or sterile cotton swab for streak inoculation), Bunsen burner, forceps (flamed for sterilisation)
- Marker pen, adhesive tape (2 pieces only — not fully sealed), ruler (mm scale), incubator

Aseptic Technique (Essential for Validity and Safety)

- Work near a **lit Bunsen burner** — the updraft of hot air creates a sterile zone by carrying airborne microorganisms upward and away from the work area
- **Flame** the neck of bottles and culture tubes before and after opening (kills organisms on the rim)

- Sterilise inoculating loops and spreaders by **flaming until red-hot**; allow to cool before use (or use pre-sterilised disposable plastic loops)
- Keep lids on Petri dishes as much as possible; do not breathe, cough, or talk over open plates
- Secure lids with **two pieces** of adhesive tape (not fully sealed — anaerobic conditions could promote growth of dangerous obligate anaerobes like *Clostridium*)

Method

1. Working near a Bunsen flame, use a sterile cotton swab or glass spreader to **spread the bacterial culture evenly** across the entire surface of the agar plate. This creates a uniform "lawn" of bacteria.
2. Using **flamed forceps**, place the antibiotic discs (or multidisc) on the agar surface. Press gently to ensure contact. Space discs evenly (at least 25 mm apart to prevent overlapping zones).
3. Seal the lid with two pieces of tape (forming an X). **Do not fully seal**. Label the base of the plate with name, date, organism, and disc identities.
4. Incubate **upside down** (prevents condensation dripping onto colonies) at **$\leq 25^{\circ}\text{C}$** for 24–48 hours.
5. After incubation, **do not open the plate**. Measure the **diameter** of each zone of inhibition in mm using a ruler held against the base of the plate. Measure at the widest point.

Safety

- Maximum incubation temperature in schools: **$\leq 25^{\circ}\text{C}$** . At 37°C (human body temperature), dangerous human pathogens could proliferate if contamination occurred.
- **Never open plates after incubation**. Unknown organisms may have grown.
- After examination, plates must be **autoclaved** (121°C , 15 min, 15 psi) before disposal in clinical waste.
- Full aseptic technique throughout to prevent contamination (both of the experiment and of the experimenter).

Analysis

- Calculate the **area** of each zone of inhibition: $A = \pi r^2$, where $r = \text{diameter}/2$. Area is more appropriate than diameter because the antimicrobial diffuses radially in two dimensions.

- Subtract the disc area if required (for a more accurate representation of the inhibited area).
- A larger zone indicates a more effective antimicrobial agent against that bacterium (or better diffusion through the agar).
- Some antibiotics are **bactericidal** (kill bacteria) and others are **bacteriostatic** (stop them growing); both produce a clear zone, so this method alone cannot tell them apart.
- Note: zone size depends on both the potency of the agent AND its ability to diffuse through agar. A highly effective agent with poor diffusion may produce a small but very clear zone.

Sources of Error and Improvements

Error	Effect	Improvement
Contamination from the environment	Extra colonies obscure zones; unreliable results	Strict aseptic technique; work near Bunsen flame; sterilise all equipment
Uneven bacterial lawn	Inconsistent zone shapes; difficult to measure accurately	Spread thoroughly and evenly using a sterile spreader; rotate the plate while spreading
Discs not in good contact with agar surface	Poor diffusion of antimicrobial; smaller or absent zones	Press each disc gently but firmly with flamed forceps to ensure full contact
Overlapping zones	Cannot measure individual zones accurately	Space discs at least 25 mm apart; use a standard template
Measuring diameter instead of calculating area	Underestimates the proportional difference between zones (area scales with r^2)	Always calculate area using $A = \pi r^2$

Exam Tip: Always calculate πr^2 for the zone area (not just diameter). Larger zone area = more effective antibiotic. Explain why incubation is at **$\leq 25^\circ\text{C}$ in schools**: to prevent the growth of potentially dangerous human pathogens (which thrive at 37°C). Explain why plates are incubated **upside down**: to prevent condensation from the lid dripping onto the

agar surface, which could cause colonies to merge and spread, making them impossible to count or measure.

CP16: Rate of Respiration (Respirometer)

Background Theory

A **respirometer** measures the rate of aerobic respiration by detecting the uptake of O_2 . Germinating seeds are used because they have a high metabolic rate. The key design feature is the inclusion of **soda lime** ($NaOH/Ca(OH)_2$), which absorbs all CO_2 produced by respiration. Since CO_2 is removed, any decrease in gas volume within the closed system is solely due to O_2 consumption, causing a coloured liquid drop in the capillary tube to move towards the organisms. A **glass bead control** tube (same volume as the seeds but with no respiration) accounts for any changes in atmospheric pressure or temperature that might independently affect gas volume.

Apparatus

- Simple respirometer (two boiling tubes or syringes connected to a capillary tube with a manometer fluid drop)
- Germinating seeds (e.g. mung beans, peas — germinated for 2–3 days), glass beads (control)
- Soda lime granules (CO_2 absorbent), cotton wool plugs (prevent soda lime contacting organisms)
- Capillary tube with scale (mm markings), coloured manometer fluid (e.g. potassium permanganate)
- Water bath (thermostatically controlled), thermometer, stopwatch

Method

1. Place a layer of soda lime at the bottom of each tube, separated from the organisms by a cotton wool plug. In one tube, place germinating seeds; in the control tube, place an equal volume of glass beads on top of soda lime (also separated by cotton wool).

2. Seal both tubes and connect them to the capillary tube containing the coloured fluid drop. Place the entire apparatus in a water bath at the desired temperature. **Equilibrate for 5 minutes.**
3. Record the starting position of the coloured fluid drop. Then record its position every minute for 10–15 minutes.
4. The drop moves **towards the organisms** as O_2 is consumed (net decrease in gas volume because CO_2 is absorbed by soda lime).
5. Repeat at different temperatures (e.g. 15, 20, 25, 30, 35, 40°C) to investigate the effect of temperature.

Safety

- **Soda lime is corrosive** (strongly alkaline) — wear chemical-resistant gloves and eye protection. Do not allow skin contact. If contact occurs, wash immediately with copious water.
- Do not allow soda lime to touch the organisms (separated by cotton wool).
- If small invertebrates (e.g. woodlice or maggots) are used instead of seeds, keep them on gauze well away from the soda lime, keep the experiment short, and return them to their habitat afterwards.
- Handle hot water baths carefully — risk of scalding.

Analysis

- Rate of O_2 uptake = distance moved by fluid drop (mm) / time (min).
- If capillary bore diameter is known, convert to volume: $V = \pi r^2 \times \text{distance}$.
- Plot **rate of O_2 uptake against temperature**. Expected: rate increases with temperature (increased kinetic energy of enzyme molecules; more frequent enzyme-substrate collisions), roughly doubling for every 10°C rise ($Q_{10} \approx 2$), until an optimum is reached. Beyond ~40°C, rate decreases as respiratory enzymes denature.
- $Q_{10} = \text{rate at } (T + 10) / \text{rate at } T$. A Q_{10} of ~2 confirms the process is enzyme-catalysed.

Sources of Error and Improvements

Error	Effect	Improvement
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Leaky apparatus (poor seals)	Air enters; fluid drop does not move reliably	Seal all joints with Vaseline or silicone grease; check for leaks before starting by watching for unexpected movement
Seeds at different germination stages	Variable metabolic rates between replicates	Use seeds from the same batch, germinated for exactly the same number of days
No control tube (glass beads)	Cannot distinguish O ₂ uptake from atmospheric pressure/temperature changes	Always include the glass bead control; subtract control movement from experimental reading
Soda lime exhausted (saturated with CO ₂)	CO ₂ no longer absorbed; volume change no longer reflects O ₂ uptake alone	Use fresh soda lime for each experiment; do not run experiments for excessively long periods
Insufficient equilibration time	Temperature not uniform; initial readings unreliable	Allow at least 5 minutes for temperature equilibration before recording

Exam Tip: Explain why the soda lime is essential: it absorbs CO₂ produced by respiration, so the only gas change is O₂ consumption, causing a net decrease in volume. Without soda lime, the CO₂ produced would partially replace the O₂ consumed (RQ $\approx$ 1 for glucose), and the fluid drop would barely move. The glass bead control accounts for changes in atmospheric pressure and temperature. State $Q_{10} = \text{rate at } (T+10) / \text{rate at } T$. If $Q_{10} \approx 2$, the process is enzyme-catalysed.

CP17: Effects of Exercise on Tidal Volume, Breathing Rate, Ventilation and Oxygen Consumption (Spirometer)

Background Theory

During exercise, muscles contract more frequently, increasing the rate of aerobic respiration: $\text{glucose} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{ATP}$. This elevates blood CO_2 concentration and lowers blood pH ($\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3 \rightarrow \text{H}^+ + \text{HCO}_3^-$). **Central chemoreceptors** in the medulla oblongata and **peripheral chemoreceptors** in the carotid and aortic bodies detect the rise in CO_2/H^+ and send nerve impulses to the respiratory centre, which increases both **breathing rate** (breaths per minute) and **tidal volume** (volume per breath). **Minute ventilation** = tidal volume x breathing rate, and this increases to supply more O_2 and remove more CO_2 .

Apparatus

- Spirometer (containing fresh soda lime and filled with medical-grade oxygen) connected to a data logger or kymograph to produce a trace; clean disposable mouthpieces or sterilised mouthpieces (one per participant)
- Stopwatch, step bench or exercise bike, metronome (to standardise step rate)
- Nose clip (essential for spirometry to ensure all air passes through the mouthpiece)
- Questionnaire for participant screening (asthma, heart conditions, recent illness)

Method

1. **Screen participants** for respiratory or cardiac conditions before beginning. Obtain informed consent.
2. Measure **resting breathing rate** (count breaths per minute for 60 s) and **resting tidal volume** (using spirometer with nose clip).
3. Subject performs standardised exercise (e.g. step-ups at 30 steps per minute for 5 minutes, using a metronome for consistency).
4. **Immediately after** exercise ceases, measure breathing rate and tidal volume again.
5. Continue measuring at **1-minute intervals during recovery** for 5–10 minutes until values return to baseline.
6. Calculate **minute ventilation** = tidal volume x breathing rate at each time point.

Reading a Spirometer Trace

- **Tidal volume** = the height of one breath on the trace (from the lowest point of a trough to the top of the next peak) during normal breathing.
- **Breathing rate** = the number of peaks per minute, read using the time axis of the trace.
- **Minute ventilation** = tidal volume $\times$ breathing rate ($\text{dm}^3 \text{ min}^{-1}$).
- **Oxygen consumption**: the soda lime absorbs the CO_2 breathed out, so the total volume of gas in the spirometer falls as O_2 is used. The trace therefore slopes downwards. Draw a line through the troughs (or peaks) and find the fall in volume per minute — this is the rate of oxygen consumption. Compare the slope at rest with the slope after exercise.

Safety

- Screen participants for respiratory conditions (asthma), cardiac problems, and injuries.
- Use clean, individual disposable mouthpieces (hygiene — prevent cross-infection).
- Use fresh soda lime and medical-grade oxygen in the spirometer; limit each recording to a few minutes and stop if the subject feels dizzy or short of breath.
- Stop exercise immediately if the participant feels dizzy, faint, experiences chest pain, or feels unwell.
- Have a first-aid kit available; work in a well-ventilated room.

Analysis

- Plot breathing rate and tidal volume as separate graphs against time (before, during, and after exercise).
- Calculate and plot minute ventilation against time.
- Both breathing rate and tidal volume increase during exercise. Recovery time (time to return to resting values) can be measured and compared between individuals.
- Fitter individuals typically have a shorter recovery time, larger tidal volumes, and lower resting heart rates.

Sources of Error and Improvements

Error	Effect	Improvement
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Inconsistent exercise intensity between participants	Cannot compare results fairly	Use a metronome to standardise step rate; use the same step height and duration for all
Air escaping through nose during spirometry	Tidal volume measurement underestimated	Use a nose clip; ensure proper mouthpiece seal
Individual variation in fitness levels	Large between-subject variation makes trends hard to see	Compare within the same individual (before vs after); use large sample sizes; calculate means
Delay between stopping exercise and first measurement	Recovery has already begun; peak values missed	Start measurements immediately (<5 seconds) after exercise stops

Exam Tip: Minute ventilation = tidal volume x breathing rate. During exercise, CO₂ concentration in the blood rises, which is detected by chemoreceptors (central in the medulla, peripheral in the carotid/aortic bodies). These send increased nerve impulses to the respiratory centre in the medulla, which increases the rate and depth of breathing. This is an example of **negative feedback** maintaining blood gas homeostasis.

CP18: Habituation to a Stimulus

Background Theory

Habituation is a simple form of learning in which an animal's response to a stimulus decreases when the stimulus is repeated and turns out to be harmless. It stops the animal wasting time and energy responding to unimportant stimuli, so it can carry on feeding or moving. A common school investigation uses a **snail**: when its head is touched, it withdraws its tentacles, and they re-emerge after a short time. With repeated gentle touches, the tentacles are withdrawn less and re-emerge sooner. At the level of the synapse, repeated stimulation means fewer calcium ion

channels open in the presynaptic membrane of the sensory neurone, so less neurotransmitter is released and fewer action potentials are produced in the motor neurone that causes withdrawal.

Apparatus

- Land snails (e.g. garden snails) kept in a moist container with food
- Clean, flat, damp surface (e.g. a glass plate or tile), cotton buds, water
- Stopwatch, recording table

Method

1. Place a snail on a damp, clean surface and leave it undisturbed for a few minutes until it is moving with its tentacles fully extended.
2. Touch the snail firmly but gently between its upper tentacles with a damp cotton bud, and start the stopwatch at the same moment.
3. Record the time taken for the tentacles to re-emerge **fully**.
4. Repeat the touch at the same place, with the same pressure, as soon as the tentacles are fully extended again (or at a fixed interval, e.g. every 30 seconds).
5. Repeat for 10–15 touches in total.
6. Repeat with other snails, allowing each one to rest between trials.

Key Variables

Variable	Detail
Independent	Number of times the stimulus has been applied
Dependent	Time for tentacles to re-emerge fully
Control variables	Position and strength of touch, interval between touches, temperature, humidity, light, noise and vibration, size and species of snail

Safety and Ethical Considerations

- Handle snails gently; keep them and the surface moist so they do not dry out; use only a light touch that does not cause harm.

- Use the fewest snails needed and return them to their habitat or tank afterwards.
- Wash hands after handling snails.

Analysis

- Plot the mean time for tentacles to re-emerge against the number of touches. Expect the time to fall as the stimulus is repeated.
- Use a statistical test, e.g. Spearman's rank correlation between touch number and re-emergence time, to judge whether the trend is significant.
- Explain the trend as habituation, and link it to reduced neurotransmitter release at synapses in the reflex pathway.
- Applying a new, different stimulus after habituation usually restores the full response, showing that the snail is not simply tired.

Sources of Error and Improvements

Error	Effect	Improvement
Strength of touch varies	Response changes for reasons other than habituation	Use a standard method, e.g. a dropper releasing a drop of water from a fixed height, or a bristle that bends at a set force
Judging when tentacles are "fully" out	Timing is subjective	Agree a clear end-point; have the same person time every trial; video the snail
Other disturbances (vibration, shadows, noise)	Extra stimuli cause withdrawal	Work on a stable bench in constant light, away from movement
Individual variation between snails	Large spread in the data	Test several snails and calculate means; use snails of similar size

Exam Tip: Define habituation clearly: a **reduced response to a repeated, harmless stimulus**. Its value is that the animal does not waste energy on responses that are not needed. You may be asked to explain it at the synapse (fewer Ca^{2+} channels open, less

neurotransmitter released), to identify variables that must be controlled, or to discuss the ethics of using animals.